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COMPREHENSIVE STUDY GUIDE

Pharmaceutical Preparations & Calculations

PCT 202 · Solutions, Phase Equilibria, Solid Dosage Forms & Calculations

PART ONE · Pharmaceutical Solutions

Dispensing · formulation · solubility · colligative properties
Distribution law · syrups, elixirs & linctuses · topical & body-cavity solutions

PART TWO · Phase Equilibria

Phase rule · one- & two-component systems · eutectic mixtures · freeze-drying

PART THREE · Powders, Tablets & Capsules

Powders & granules · granulation · tablet types · hard & soft capsules

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Units · ratio & proportion · specific gravity · percentage strength · alligation

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§1 DISPENSING & THE NATURE OF SOLUTIONS

Dispensing is the branch of pharmaceuticals concerned with mixing, preparing, presenting and supplying medicines to the patient in a form that can actually be used — solutions, suspensions, emulsions, tablets, capsules, suppositories, ointments, creams and so on. Sound dispensing therefore rests on a working knowledge of how a preparation is made, labelled, packaged, stored and dosed, and of how stable it will remain in the patient's hands.

A pharmaceutical solution is a **homogeneous, single-phase system** in which one or more solutes are dissolved in a solvent. The solvent — almost always a liquid — is the continuous phase in which dispersion happens; the solute, which may be a solid, a liquid or a gas, is broken down into individual ions or molecules and distributed evenly throughout. In most preparations the solvent is present in far greater quantity than the solute. A notable exception is Syrup BP, in which sucrose (the solute) makes up about 66.7% and the solvent only about 33.3%.

KEY POINT

Because a true solution is molecularly homogeneous, every measured portion contains the same amount of drug. This built-in dose uniformity is one of the strongest reasons for choosing a solution.

§2 FORMULATING A SOLUTION

Turning a drug into a palatable, stable liquid usually calls for several classes of additive besides the active ingredient and the vehicle. The table below groups the excipients most often required and why each is included.

Component	Role in the formulation	Typical examples
Solvent / vehicle	Dissolves the drug; may be aqueous or non-aqueous	Water, aromatic waters, alcohol, fixed oils, liquid paraffin
Co-solvents	Improve solubility of poorly soluble drugs	Alcohol, propylene glycol, glycerol
Buffers	Hold the preparation within a required pH range	Carbonates, citrates, phosphates
Sweeteners	Improve palatability	Sucrose; sorbitol or mannitol for diabetics
Flavours	Mask taste / odour	Anise, raspberry, peppermint, lemon; synthetic esters and aldehydes
Colourants	Aid identification and appearance	Water-soluble dyes such as amaranth, tartrazine
Preservatives	Prevent microbial spoilage	Benzoic acid, phenol, benzalkonium chloride
Antioxidants	Prevent oxidative degradation	Propyl esters, tocopherol
Viscosity enhancers	Adjust flow and mouth-feel	Hydroxyethylcellulose, povidone

The choice of vehicle is guided by clarity, viscosity, toxicity, odour, compatibility, palatability and colour. Antioxidants matter particularly where vitamins or essential oils can be oxidised, a reaction that light or trace metals such as copper or iron may initiate.

§3 ADVANTAGES, DISADVANTAGES & USES

Advantages

- » Easier to swallow than tablets or capsules, which makes them well suited to paediatric and geriatric patients.
- » Faster onset of action: the drug is already dissolved, so it bypasses the disintegration and dissolution steps a solid must undergo before absorption.
- » Uniform dosing — the drug stays evenly distributed, avoiding the dose variation seen in some solid and disperse systems.
- » Gentler on the gastric mucosa, reducing the irritation caused by concentrated solid particles of irritant drugs.
- » Flexible dosing and generally simple to prepare; an attractive appearance can also carry a useful psychological effect.

■ Disadvantages

- » Bulky and heavy, so they are awkward to transport and store.
- » A broken container means the entire dose is lost at once.
- » Chemically less stable than solids, giving a shorter shelf-life.
- » The aqueous vehicle encourages microbial growth, so preservatives are usually needed.
- » Unpleasant tastes become more obvious in solution, forcing the use of sweetening and flavouring agents.
- » Accurate measurement depends on a proper spoon or device and some technique on the patient's part.

■ Uses of pharmaceutical solutions

Solutions are formulated for many routes of administration:

- » **Oral** — mixtures, elixirs, linctuses, draughts, syrups and paediatric drops.
- » **Mouth and throat** — mouthwashes, gargles, throat paints and throat sprays.
- » **Dermatological / external** — collodions, liniments, lotions and paints.
- » **Body cavities** — douches, enemas, ear drops and nasal drops or sprays.
- » **Parenteral** — intramuscular, intravenous and subcutaneous injections.

§4 SOLUBILITY & TYPES OF SOLUTION

Solubility is the amount of a substance that will dissolve in a stated amount of solvent to form a saturated solution at a given temperature. Understanding it matters even for non-solution dosage forms, because any drug must eventually dissolve in body fluids before it can be absorbed. Solutions can involve any combination of states — solid-in-solid (alloys such as solder), solid-in-liquid (sucrose in water), solid-in-gas (iodine vapour in air), liquid-in-liquid (mineral oil in paraffin), liquid-in-gas (water vapour in air) and gas-in-gas (air). The solid-in-liquid type is by far the most important in pharmacy.

■ Types of solution by degree of saturation

Type	Description
Saturated	In equilibrium with excess solute: molecules leave and re-enter the solid at equal rates at the given temperature.
Unsaturated (sub-saturated)	Contains less dissolved solute than is needed for saturation at that temperature.
Supersaturated	Contains more dissolved solute than saturation allows — a metastable state. Seeding with a crystal, vigorous shaking or scratching the container makes the excess crystallise out.

Sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$) and sodium acetate are classic examples that dissolve fully when hot and then fail to crystallise on cooling, giving a supersaturated solution.

■ Expressing solubility

In pharmacy three concentration expressions dominate:

- » % w/w — grams of solute per 100 g of solution.
- » % w/v — grams of solute per 100 mL of solution.
- » % v/v — millilitres of solute per 100 mL of solution.

Where an exact figure is unknown or unhelpful, pharmacopoeias use descriptive terms defined by the parts of solvent needed to dissolve one part of solute:

Descriptive term	Parts of solvent per 1 part of solute
Very soluble	less than 1
Freely soluble	1 – 10
Soluble	10 – 30
Sparingly soluble	30 – 100
Slightly soluble	100 – 1 000
Very slightly soluble	1 000 – 10 000
Practically insoluble / insoluble	greater than 10 000

§5 FACTORS AFFECTING SOLUBILITY

1. Temperature

For a solute that dissolves endothermically (absorbing heat), solubility rises as temperature rises. For an exothermic dissolution (releasing heat) the opposite holds. Calcium citrate, calcium hydroxide, paraldehyde and most gases dissolve exothermically, so their solubility falls as temperature climbs.

2. Particle size

Reducing particle size increases surface area and therefore solubility and dissolution rate. The solubility of barium sulphate in water roughly doubles when particle size falls from 1.8 μm to 0.1 μm , and micronising griseofulvin markedly increases its activity.

3. Pressure

Pressure has little effect on solids and liquids but a strong effect on gases. For a gas, higher pressure increases solubility according to Henry's law, $M = kP$ at constant temperature, where M is the mass of gas dissolved per unit volume of solvent at pressure P and k is a constant.

4. Nature of the solvent

The solvent's own character governs how much solid it can hold, and adding other substances can change a solvent's properties and hence the solubility of a solute in the system.

5. pH

A weakly acidic drug such as aspirin is more soluble in alkaline media, where it forms a soluble salt, but precipitates in acid. A weak base behaves in the opposite way.

§6 COLLIGATIVE PROPERTIES OF SOLUTIONS

Colligative properties depend only on the **number** of solute particles present, not on their chemical nature or size. For a non-volatile solute in a solvent there are four: lowering of vapour pressure, elevation of boiling point, depression of freezing point and osmotic pressure.

6.1 Lowering of vapour pressure

When a non-volatile solute dissolves in a solvent, solute particles at the surface obstruct the escape of solvent molecules, so fewer escape and the vapour pressure falls. Raoult's law states that the vapour pressure of solvent above a solution is proportional to the mole fraction of the solvent. Writing X_1 and X_2 for the mole fractions of solvent and solute ($X_1 + X_2 = 1$):

RELATIVE LOWERING

$$P = P_1^\circ \cdot X_1 = P_1^\circ (1 - X_2)$$

$$\Delta P = P_1^\circ - P = P_1^\circ \cdot X_2$$

$$\therefore \Delta P / P_1^\circ = X_2 = n_2 / (n_1 + n_2)$$

So the *relative* vapour-pressure lowering equals the mole fraction of solute — a colligative quantity that depends only on how many solute particles are present.

WORKED EXAMPLE — 171.2 g sucrose in 1000 g water at 20°C ($P^\circ = 17.535$ mm)

$$n_1 (\text{water}) = 1000 / 18.02 = 55.5 \text{ mol}$$

$$n_2 (\text{sucrose}) = 171.2 / 342.3 = 0.500 \text{ mol}$$

$$(a) \Delta P / P^\circ = 0.500 / (55.5 + 0.50) = 0.0089$$

$$(b) \Delta P = 0.0089 \times 17.535 = 0.156 \text{ mm}$$

$$(c) P = 17.535 - 0.156 = 17.379 \text{ mm}$$

6.2 Elevation of boiling point

The normal boiling point is where vapour pressure equals the external pressure (760 mmHg). Because a non-volatile solute lowers the solvent's vapour pressure, a higher temperature is needed to reach 760 mmHg, so the boiling point rises. The elevation is $\Delta T_b = K_b m$, where K_b is the molal (ebullioscopic) constant characteristic of the solvent and m is molality.

WORKED EXAMPLE — 0.200 m solution, $\Delta T_b = 0.103^\circ$

$$K_b = \Delta T_b / m = 0.103 / 0.200 = 0.515$$

6.3 Depression of freezing point

Dissolving a solute lowers the escaping tendency of the solvent, so the temperature must fall further before solid and liquid reach equilibrium; the freezing point of a solution is therefore always below that of the pure solvent. The depression is $\Delta T_f = K_f m = K_f (1000 w_2) / (w_1 M_2)$, where K_f is the cryoscopic constant.

WORKED EXAMPLES

$$3.42 \text{ g sucrose in } 500 \text{ g water } (K_f = 1.86):$$

$$\Delta T_f = 1.86 \times (1000 \times 3.42) / (500 \times 342) = 0.036 \Rightarrow -0.036^\circ\text{C}$$

$$1.3 \text{ m sucrose } (K_f = 2.1): \Delta T_f = 2.1 \times 1.3 = 2.73^\circ\text{C}$$

6.4 Osmotic pressure

Osmosis is the movement of solvent through a semi-permeable membrane from a region of lower solute concentration to one of higher solute concentration. The **osmotic pressure** is the external pressure that must be applied to the more concentrated solution to stop solvent flowing into it. It underpins the design of isotonic parenteral and ophthalmic products.

HOW THEY ARE MEASURED

Vapour-pressure lowering — manometer or the isopiestic method; boiling-point elevation — the Cottrell apparatus; freezing-point depression — the Beckmann, Rast (camphor) or equilibrium methods.

§7 SOLUTIONS OF LIQUIDS IN LIQUIDS

A liquid-in-liquid solution is a homogeneous mixture of one liquid dispersed in another. Such pairs are described as **miscible** when they blend completely or **immiscible** when they separate into layers.

What decides miscibility

Like dissolves like

Liquids of similar polarity and structure mix freely: polar with polar, non-polar with non-polar. Benzene and toluene, n-hexane and n-heptane, and ethyl bromide and ethyl iodide are all miscible pairs. A non-polar liquid such as oil forms two layers with water, whereas a polar liquid such as ethanol blends completely; and a non-polar solid like iodine dissolves in non-polar lighter fluid but not in water.

Intermolecular forces

Liquids are fully miscible when their intermolecular forces are alike. Water mixes with other hydrogen-bonding liquids, while non-polar hydrocarbons rely on weak London dispersion forces. Alcohols $\text{CH}_3(\text{CH}_2)_n\text{OH}$ are amphipathic — a hydrophilic $-\text{OH}$ end and a hydrophobic chain. When n is large the hydrocarbon part dominates and water solubility is limited; when n is very small the $-\text{OH}$ group dominates, so methanol ($n = 0$) and ethanol ($n = 1$) are fully miscible with water but less so with hydrocarbons.

Temperature

Raising the temperature can increase the mutual miscibility of two partially miscible liquids.

Miscible examples	Immiscible examples
Ethanol in water (alcoholic beverages)	Oil and water
Hydrochloric acid in water	Benzyl alcohol and water
Gasoline (blended hydrocarbons)	Benzene and water

§8 DISTRIBUTION BETWEEN IMMISCIBLE LIQUIDS

In 1891 Walther Nernst formulated the **distribution (partition) law**: when a solute that is soluble in each of two immiscible liquids is shaken with both, it distributes itself so that the ratio of its concentrations in the two phases is constant at a fixed temperature. That constant is the partition or distribution coefficient, K_{pc} (or K_{d}) — a dimensionless equilibrium constant, usually evaluated from solubilities (g/100 mL) or molar concentrations.

CONDITIONS FOR NERNST'S LAW

The law holds only when: the temperature is constant; both solutions are dilute; the solute does not react with either solvent; and the solute exists in the same molecular state in both phases (no association or dissociation).

■ Applications in pharmacy

- » **Solvent extraction** — separating a drug from a mixture using a selective solvent.
- » **Chromatography** — a solute partitions repeatedly between a mobile phase and an immiscible stationary phase.
- » **Formulation** — predicting drug solubility across solvents when designing solutions, suspensions and emulsions.
- » **Drug absorption** — a drug must partition from the aqueous gut contents into the lipid membrane and back into blood; partitioning describes this movement between lipid and water.
- » **Drug distribution** — after absorption a drug partitions from blood into tissues according to its affinity for each compartment.

§9 ORAL SOLUTIONS — SYRUPS, LINCTUSES & ELIXIRS

■ 9.1 Syrups

Syrups are concentrated, viscous solutions of sugar in water or another aqueous liquid. A **simple syrup** contains up to about 85% sucrose (a minimum near 65% is needed for self-preservation). A **medicinal syrup** is simple syrup plus a therapeutic agent — for example salbutamol (bronchodilator), cefuroxime (a second-generation cephalosporin), bromhexine (mucolytic) or chlorpheniramine (antihistamine). A **flavoured syrup** is simple syrup plus a flavour, used mainly as a vehicle for extemporaneous products.

Methods of preparation

- » **Solution with heat** — for constituents that are neither volatile nor heat-sensitive, and when speed is needed; the syrup is strained to remove impurities and made up to volume.
- » **Agitation without heat** — for heat-labile or volatile constituents; prepared in a bottle about twice the required size to allow vigorous shaking.
- » **Addition of medicating liquid** — fluid extract is mixed with water, left to let insoluble matter separate, filtered, and the sucrose then dissolved in the filtrate.
- » **Percolation** — purified water is passed slowly through a bed of sucrose in a percolator, the flow controlled by a stop-cock.

General method for preparing a solution

Reduce particle size if needed (mortar and pestle, then rinse it out), calibrate a suitable container, weigh and transfer the ingredients, stir to dissolve, add the other liquid ingredients (holding back anything volatile), transfer to the pre-calibrated bottle, add the volatile excipient last, rinse the beaker into the bottle, make up to volume, then clean and label appropriately.

■ 9.2 Linctuses

A linctus is a viscous oral solution intended to relieve cough by acting mainly in the throat. It contains a high proportion of syrup and glycerol for their demulcent (soothing) effect, the dose volume is small (about 5 mL), and it is labelled “to be swallowed slowly without the addition of water.” Active ingredients are typically sedatives and expectorants. Common vehicles are syrup (sweet, viscous, preservative), glycerol (demulcent, sweet), sorbitol (for diabetics) and invert sugar.

Typical adjuncts include a chemical stabiliser (invert sugar, which is reducing and limits oxidation), colourants (amaranth, tartrazine), flavours (lemon, blackcurrant, benzaldehyde spirit) and preservatives.

EXAMPLE — Codeine Linctus, Paediatric BPC (per 1000 mL)

Codeine phosphate 3 g · Lemon syrup 200 mL · Benzoic acid soln 20 mL

Chloroform spirit 20 mL · Tartrazine soln 10 mL · Syrup to 1000 mL

Method: dissolve codeine phosphate in warm water; add colour, benzoic acid and chloroform spirit one at a time with mixing; make up to volume with syrup.

Use: cough suppressant.

9.3 Elixirs

Elixirs are clear, pleasantly flavoured and usually attractively coloured oral solutions of potent drugs (antibiotics, antihistamines and the like) that act mainly in the gastrointestinal tract. To keep essential oils in solution, 10–20% alcohol is often added, and mixed solvents are used to dissolve the drug — phenobarbitone in alcohol, glycerol and water; paracetamol in alcohol, propylene glycol and glycerol. Alcohol is avoided in paediatric elixirs, where propylene glycol is the main co-solvent. Elixirs are less viscous and less sweet than linctuses, and their alcohol content also preserves them.

EXAMPLE — Piperazine Citrate Elixir (per 1000 mL)

Piperazine citrate 180 g · Chloroform spirit 5 mL · Glycerin 100 mL

Orange oil 0.25 mL · Syrup 500 mL · Purified water to 1000 mL

Method: dissolve piperazine citrate in a little water; mix in orange oil, glycerin, syrup and chloroform spirit; make up to final volume. Use: anthelmintic.

9.4 Syrup vs elixir vs linctus

Parameter	Syrup	Elixir
Alcohol	Not necessary	Necessary
Sweetness	More	Less
Viscosity	More	Less
Sugar content	High	Lower
Stability	Less	More
Taste masking	More effective	Less effective
Clarity	May not be clear	Clear solution
Diabetic patients	Sucrose syrup unsuitable	Readily used

Parameter	Linctus	Elixir
Sweetness	More	Less
Viscosity	More	Less
Alcohol	Not necessary	Necessary
Site of action	Throat	Gastrointestinal tract
Typical actives	Expectorants, sedatives	Antibiotics, antihistamines

§10 SOLUTIONS FOR BODY CAVITIES

■ Paints

Paints contain one or more actives applied with a brush to the skin or a mucous membrane, usually for their antiseptic or local-analgesic action. Skin paints use a volatile vehicle such as ethanol that evaporates quickly to leave a film of drug; throat and other mucosal paints use a viscous vehicle such as glycerol to hold the drug at the site.

EXAMPLE — Compound Iodine Paint (Mandl's Paint), for pharyngitis / tonsillitis

Potassium iodide 25 g · Iodine 12.5 g · Alcohol 40 mL · Water 25 mL

Peppermint oil 4 mL · Glycerol to 1000 mL

Dispense in an air-tight, coloured, narrow-mouthed bottle with a brush.

Label: for external use; not to be swallowed in large quantities. Store cool and dark.

■ Gargles and mouthwashes

Gargles are concentrated aqueous solutions diluted with warm water before use to prevent or treat throat infections; antiseptics are often included, though usually below a truly antibacterial concentration. The liquid is held in the throat and agitated by exhaling through it, then spat out. **Mouthwashes** are similar concentrated aqueous solutions with deodorant, antiseptic, analgesic or astringent properties, used on the lining of the mouth rather than the throat. Typical ingredients include a vehicle (water), flavour (menthol), sweetener (saccharin, sucrose), preservative (sodium benzoate), antiseptic (fluoride) and a detergent (sodium lauryl sulphate).

EXAMPLE — Compound Sodium Chloride Mouthwash (per 1000 mL)

Sodium bicarbonate 10 g · Sodium chloride 15 g · Chloroform water 500 mL

Peppermint emulsion 25 mL · Water to 1000 mL

Method: dissolve the salts in some water; add chloroform water and peppermint emulsion; make up to volume. Label: not to be swallowed in large quantity.

■ Eye, ear and nasal preparations

- » **Eye preparations** — for lubrication, cleaning, treatment or hydrating contact lenses; must be sterile and attention paid to tonicity (hyper-, hypo- or isotonic).
- » **Ear (otic/aural) preparations** — drops, sprays or washes for the external auditory canal in water, propylene glycol, glycerol or alcohol/water; used as antibiotics, antiseptics, cleansers or wax softeners.
- » **Nasal solutions** — small-volume drops or sprays, isotonic and around pH 6.8; extreme pH damages the cilia.
- » **Enemas (rectal) and douches (nasal or vaginal)** — for cleansing, treatment or diagnosis; large volumes are warmed to body temperature before use.

§11 TOPICAL SOLUTIONS

■ Collodions

A collodion is a solution of pyroxylin (nitrocellulose) in a mixture of organic solvents, typically ether or ethanol. Painted onto the skin, the solvent evaporates to leave a flexible film; castor oil in the formula confers the flexibility. A drug such as salicylic acid can be dissolved in the base to make a **medicated collodion**. Collodions seal minor cuts and wounds or hold a medicament against the skin for long periods.

CAUTION

Collodions are highly inflammable. Dispense in small (5–10 mL) amber, tightly closed containers to prevent solvent loss and concentration change; label “for external use only, highly inflammable, keep away from naked flames and out of reach of children.”

Lotions

Lotions are most often solutions (though they may be suspensions or emulsions) applied without friction to unbroken skin as antiseptics, antipruritics, astringents or protectives. After the solvent evaporates they leave a thin film of medicament; the evaporation itself cools and can ease inflammation — ethanol accentuates the cooling while glycerol keeps the skin moist. Typical medicaments include zinc oxide (antiseptic/screening), calamine (astringent) and benzocaine (anaesthetic). Dispense in a well-closed, narrow-mouthed, amber fluted bottle; label “for external use only, shake well, store cool.”

Liniments

Liniments (once called embrocations) are applied with friction to unbroken skin to ease sprains and strains, and may be solutions or emulsions. **Alcoholic liniments** are rubefacient, counter-irritant and mildly astringent with good penetration; **oily liniments** are milder and better for massage. Label “for external use only, not to be applied to broken skin.”

Irrigation solutions

Irrigation solutions wash surgical incisions and wounds. Because they contact exposed tissue they must be sterile and pyrogen-free, and are commonly prepared extemporaneously as isotonic sodium chloride.

Liniment vs lotion

Parameter	Liniment	Lotion
Absorption	Faster	Slower
Viscosity	Low	Higher (especially as suspension)
Cosmetic appearance	Less	More
Purpose	Pain, inflammation	Protection / antiseptic
Application	With friction	Without friction

Parenteral solutions deserve a final mention: their design must weigh the route of administration, the volume injected, tonicity, whether a preservative is appropriate, and above all sterility.

§12 PHASES, EQUILIBRIUM & THE PHASE RULE

A **phase** is a homogeneous part of a system with uniform physical and chemical properties; it may be gas, liquid or solid. Several phases can coexist, each with a distinct, separable boundary — a pure compound such as water can exist as solid, liquid and gas. A system is at **equilibrium** when its free energy is at a minimum ($\Delta G = 0$) and its properties no longer change with time, with temperature, pressure and volume all constant. **Phase equilibrium** is simply the reversible balance reached when phases coexist — though only for a limited time, as ice lasts far longer in cold water than in boiling water.

The **Gibbs phase rule** gives the smallest number of independent variables (temperature, pressure, concentration) that can be changed without altering the equilibrium — that is, without creating or destroying a phase:

GIBBS PHASE RULE

$$F = C - P + 2$$

F = degrees of freedom · C = number of components · P = number of phases

KEY RELATIONSHIP

The more phases in equilibrium, the fewer the degrees of freedom. A phase diagram (P–T diagram) is the graphical picture of these equilibria.

§13 ONE-COMPONENT PHASE DIAGRAMS

For a one-component system such as water, the P–T diagram is divided by three curves meeting at a single point:

Feature	Equilibrium it represents
Curve OC (fusion / melting curve)	Solid ■ liquid (ice and water)
Curve OA (vaporisation curve)	Liquid ■ vapour
Curve OB (sublimation curve)	Solid ■ vapour
Triple point (O)	All three phases coexist at one fixed T and P
Areas between the curves	Single-phase regions — ice, water, or water vapour

■ Applying the phase rule to water

Situation	Calculation	F	Meaning
One phase (an area)	$1 - 1 + 2$	2	Bivariant — T and P vary independently
Two phases (a curve)	$1 - 2 + 2$	1	Univariant — fixing T fixes P automatically
Three phases (the point)	$1 - 3 + 2$	0	Invariant — the triple point, fixed T and P

For water the triple point occurs at $T = 0.0098^\circ\text{C}$ and $P = 4.58 \text{ mmHg}$; any change in temperature or pressure alters the number of phases present. A single phase is shown as an area, a two-phase equilibrium as a line, and the three-phase equilibrium as a single point.

§14 TWO-COMPONENT SYSTEMS & EUTECTICS

When two components are mixed, the count of components and phases changes. A solution of sodium chloride in water is a **one-phase, two-component** system, for which $F = 2 - 1 + 2 = 3$ — the three variables being temperature, pressure and concentration. Once excess solid sodium chloride is present, the system has **two components and two phases** (aqueous solution plus solid salt): a condensed system that behaves as a eutectic.

EUTECTIC MIXTURE

A eutectic mixture is a homogeneous mixture of two or more components that do not react to form a new compound but melt or solidify at a single temperature lower than the melting point of any pure component. Sodium chloride and water form such a mixture.

§15 SUBLIMATION & FREEZE-DRYING

Sublimation is the direct transition of a solid to a gas without passing through the liquid state. For water it occurs below the triple point and, unlike evaporation, involves no input of heat. This is the basis of **freeze-drying (lyophilisation)**, which preserves a sample by freezing it and then removing the ice as vapour by sublimation, giving a high-quality product that rehydrates rapidly and almost completely. Sublimation of ice is only possible below the eutectic point, and freeze-drying is always carried out under vacuum, the sublimation temperature being set by the reduced pressure.

■ The three stages

- » **Freezing** — the sample is frozen so its water becomes ice. Rapid freezing gives small crystals that damage structure little but leave fine pores that resist vapour flow; slow freezing gives large crystals that dry faster but can damage cells and denature protein.
- » **Primary drying (sublimation)** — the ice vaporises directly to vapour, which is collected on condensers and re-frozen once outside the chamber.
- » **Secondary drying (desiccation)** — residual water that never froze is removed.

Advantages	Disadvantages
Minimal damage to heat-labile materials	High capital cost of equipment
Produces a porous, friable structure	High energy cost
Rapid, near-complete rehydration	Lengthy process time
Little loss of activity	Possible pH / tonicity damage as solutes concentrate

Because of these costs, freeze-drying is reserved for high-value products — microbial cultures, coffee and tea, egg and milk products, fruits and juices, some vegetables, and injections that are unstable in solution and so are supplied as powders for reconstitution just before use.

§16 POWDERS AS A DOSAGE FORM

A **powder** is a dry solid made of many finely divided particles (roughly 10 μm to 1000 μm) produced by crushing, grinding or comminution. Powders have a large specific surface area and surface free energy, and properties such as particle size, shape, density, porosity and flowability influence how well they form, pack and process into granules, tablets, capsules and suspensions, and can affect a drug's solubility and bioavailability. Powders serve both as excipients (diluents, disintegrants, lubricants) and as a dosage form in their own right.

As a dosage form a “powder” is a mixture of powdered drug and excipients. Their use has declined as tablets and capsules have taken over, but they retain some advantages.

Advantages of powders	Disadvantages of powders
Chemically stable	Inconvenient to carry compared with tablets/capsules
Convenient for large doses	Unpleasant tastes are hard to mask
Faster dissolution of soluble drugs (no shell or tablet to disintegrate first)	Not suitable for potent low-dose drugs
	Unsuitable for drugs that irritate or damage the stomach

■ 16.1 Bulk powders

Bulk powders are a mixture of all the materials packed into a single container such as a glass or plastic bottle. Their main weakness is dose inaccuracy — the measured dose depends on the spoon used, storage humidity, settling and patient technique — so they suit drugs with a wide therapeutic window, a large dose and a pleasant taste. **Effervescent powders** are a special type containing an effervescent couple (sodium bicarbonate and citric acid) that fizzes on contact with water and helps mask salty or bitter tastes.

Bulk powders for external use include:

- » **Dusting powders** — micronised powders for local, non-absorbed therapeutic, lubricant or protective use; need excellent flow and are dusted from a perforated-lid container or a pressure aerosol.
- » **Insufflations and dry-powder inhalers (DPIs)** — fine powders delivered to the nose, ear or throat, or to the lung for asthma and COPD; often the drug is blended with a carrier such as lactose in a capsule that the device ruptures on use.

■ 16.2 Divided powders

Divided powders are bulk powders in which each individual dose is packed separately — traditionally in wrapped paper, but modern foil and plastic laminates protect volatile, hygroscopic or deliquescent materials far better and allow effervescent powders to be packed as single doses.

§17 GRANULES & GRANULATION

Because powders of differing particle size can segregate in hoppers or on storage, giving a non-uniform product, mixed powders are often granulated. **Granules** are dry aggregates of powder particles, strong enough to withstand handling, taken orally either as they are, chewed, or dispersed in liquid. They typically measure 0.2–0.5 mm as a manufacturing intermediate and 1–4 mm as a finished dosage form, and are best for low-toxicity, high-dose drugs such as methylcellulose (a bulk laxative). Categories include effervescent, coated, gastro-resistant and modified-release granules.

Granulation is the process of making discrete powder particles adhere into larger multi-particle granules. It is carried out for several reasons:

- » To prevent segregation of the constituents of the powder mix.
- » To improve the flow properties of the powder.
- » To improve compaction characteristics before tableting.
- » To reduce the dust hazard of toxic materials.
- » To limit caking of slightly hygroscopic materials, since granules absorb some moisture yet keep flowing.
- » To occupy less volume per unit weight (higher bulk density), easing storage and shipment.

Advantages of granules	Disadvantages of granules
More chemically stable than liquids (e.g. reconstitutable antibiotic syrups keep 2–3 years dry, 1–2 weeks once mixed)	Bulk granules are inconvenient to carry and self-administer
Convenient for high-dose drugs	Unpleasant tastes can be hard to mask
Faster dissolution than tablets or capsules	Not suitable for potent low-dose drugs (dosing by spoon is inaccurate)

Powders and granules are unsuitable for drugs inactivated by, or damaging to, the stomach; these are better presented as enteric-coated tablets.

§18 TABLETS

The USP defines tablets as solid dosage forms containing medicinal substances with or without suitable diluents, made by compression or by moulding. The overwhelming majority are **compressed tablets**, formed by applying high pressure through steel punches and dies to powders or granulations. Tablets may be discoid, round, oval, oblong, cylindrical, triangular or other shapes, and vary widely in size and weight. The main types are summarised below.

Tablet type	Key features
Compressed (CT)	Simplest form, no coating; made from powdered, crystalline or granular material with binders, disintegrants, lubricants, diluents and colourants.
Sugar-coated (SCT)	Compressed tablet with a coloured sugar coating that masks taste/odour and protects oxidation-sensitive drugs.
Film-coated (FCT)	Thin water-soluble polymer film; same benefits as sugar coating but much faster to apply — now the first choice.
Enteric-coated (ECT)	Coating resists gastric fluid but dissolves in the intestine; protects acid-labile drugs and the mucosa, or delays release.
Multiple-compressed (MCT)	Made by more than one compression cycle to separate incompatible actives or ensure uniform mixing.
Layered	Additional granulation compressed onto an existing layer to give two or more layers.
Controlled-release (CRT)	Release the drug slowly over a prolonged period; also called sustained-, extended-, prolonged- or delayed-release, and repeat-action / pulsatile.
Tablets for solution (CTS)	Dissolved to make a solution (e.g. halazone, potassium permanganate); labelled not to be swallowed.
Effervescent	Contain sodium bicarbonate and an organic acid that release CO ₂ in water, acting as disintegrant and producing effervescence.

Tablet type	Key features
Buccal & sublingual	Small flat tablets held in the buccal pouch (slow) or under the tongue (rapid, e.g. nitroglycerin) for direct absorption.

§19 CAPSULES

Capsules enclose the drug in a hard or soft soluble shell, traditionally of gelatin but increasingly of hydroxypropyl methylcellulose (HPMC), which holds less residual moisture — useful for hydrolytic drugs — and suits strict vegetarians. Gelatin is obtained by hydrolysis of collagen: Type A from pork skin by acid processing and Type B from bone and skin by alkaline processing, blended to give the viscosity and bloom strength needed for manufacture. Hard-gelatin capsules let the prescriber choose a single drug or a combination at the exact dose suited to the patient — a flexibility tablets lack — and many patients find capsules easier to swallow.

Capsules owe their popularity to the following:

- » They mask the taste and odour of unpleasant drugs.
- » They are attractive in appearance.
- » They are slippery when moist, so easy to swallow with water.
- » Properly stored shells resist considerable mechanical stress.
- » They need fewer additives than tablets.
- » Their finely powdered contents give rapid, uniform release in the gastrointestinal tract.
- » The shells can be opacified with titanium dioxide or coloured to protect against light.
- » The shells are physiologically inert and readily digested.
- » No complex machinery is needed to dispense a few capsules extemporaneously.

§20 FOUNDATIONS & UNITS OF MEASUREMENT

Calculation runs through every part of pharmacy — from the physical and chemical properties of drug substances, through pharmacokinetics, formulation and dosing, to pharmacoeconomics. Because each result bears directly on patient safety, accuracy is paramount, and success depends on understanding the goal of a problem, choosing the right arithmetic, and carrying it out correctly.

RULES FOR ACCURATE CALCULATION

1. Always precede a decimal point with a zero (0.5, not .5).
2. Never alter a figure — strike out an error and rewrite it beside or above.
3. Show the full procedure, stating any conversion factors used.
4. Per the USP, keep all figures through the working and round only the final result to the appropriate significant figures.

Pharmacy uses the International System of Units (the metric system). Two other systems appear occasionally: the **avoirdupois** system of commerce (ounce, pound, pint, gallon) and the largely historic **apothecaries'** system. Frequent conversions — for instance milligrams to micrograms — make the equivalents below worth memorising.

Quantity	Equivalent	Quantity	Equivalent
1 inch	2.54 cm	1 pound (avoir)	454 g
1 metre	39.37 in	1 ounce (avoir)	28.4 g
1 fluid ounce	29.57 mL	1 kilogram	2.2 lb
1 pint (16 fl oz)	473 mL	1 gram	15.4 gr
1 quart (32 fl oz)	946 mL	1 grain	64.8 mg
1 gallon (US)	3785 mL	1 millilitre	16.2 minims
1 gallon (UK)	4545 mL	1 dL	100 mL

§21 RATIO & PROPORTION

A **ratio** is the relative size of two quantities; 1:2 is read “one is to two” and behaves like the fraction $\frac{1}{2}$. Multiplying or dividing both terms by the same number leaves the value unchanged, so 20:4, 80:16 and 5:1 are equivalent ratios (each equal to 5). A **proportion** states that two ratios are equal and may be written $a:b = c:d$, $a:b :: c:d$ or $a/b = c/d$, read “a is to b as c is to d.” Here a and d are the extremes and b and c the means.

THE KEY PRINCIPLE

In any proportion the product of the extremes equals the product of the means. Given any three terms the fourth follows: if $a/b = c/d$ then $a = bc/d$, $b = ad/c$, $c = ad/b$ and $d = bc/a$. Always label the units in each position (mL, mg) so the ratios stay correctly matched.

WORKED EXAMPLE

If 3 tablets contain 975 mg of aspirin, how much is in 12 tablets?

$$3 \text{ tab} / 12 \text{ tab} = 975 \text{ mg} / x \text{ mg}$$

$$x = (12 \times 975) / 3 = 3900 \text{ mg}$$

§22 DENSITY & SPECIFIC GRAVITY

Density is mass per unit volume; for water it is 1 g/mL. **Specific gravity** is the ratio of the weight of a substance to the weight of an equal volume of a standard (water for liquids and solids) at the same temperature — a dimensionless number. A specific gravity below 1 means the substance is lighter than water; above 1 means heavier.

CORE RELATIONSHIPS

Specific gravity = weight of substance ÷ weight of equal volume of water

Weight (g) = volume (mL) × specific gravity

Volume (mL) = weight (g) ÷ specific gravity

Specific gravity may be determined from a known weight and volume, with a pycnometer (specific-gravity bottle), or by displacement (the plummet method). It is used pharmaceutically to convert between weight and volume — for instance in automated compounders preparing total parenteral nutrition, where final volume is derived from weight divided by the known specific gravity — and clinically in urinalysis, where normal urine falls between 1.010 and 1.025; a high value indicates concentrated urine (dehydration, glucosuria, proteinuria) and a low value dilute urine (diabetes insipidus, renal disease, over-hydration).

WORKED EXAMPLE

250 mL of sevoflurane has specific gravity 1.52. Find its weight.

250 mL of water weighs 250 g $\Rightarrow$ weight = $250 \times 1.52 = 380$ g

§23 PERCENTAGE STRENGTH

Percent means parts per hundred; 50%, 50/100 and 0.5 are equivalent. For calculation a percent is converted to a decimal by dividing by 100 ($12.5\% = 0.125$). Percentages both express the strength of a component and let us work out how much to use for a required strength. The USP defines three expressions:

Expression	Meaning	Default use of unqualified “%”
% w/v	g of constituent per 100 mL of solution	Solids in liquids; gases in liquids
% v/v	mL of constituent per 100 mL of solution	Liquids in liquids
% w/w	g of constituent per 100 g of preparation	Mixtures of solids or semisolids

23.1 Weight in volume (w/v)

For w/v the solution is assumed to have the specific gravity of water, so 100 mL is treated as 100 g and a 5% w/v preparation contains 5 g per 100 mL.

WORKED EXAMPLES

Dextrose for 4000 mL of 5% solution: 100 mL $\rightarrow$ 5 g, so 4000 mL $\rightarrow$ 200 g.

0.1 mg travoprost in 2.5 mL: $(0.0001 \text{ g} \times 100) / 2.5 = 0.004\%$ w/v.

23.2 Weight in weight (w/w)

For w/w the number of grams of active is expressed per 100 g of preparation; when a specific gravity is given, convert the stated volume to weight first.

WORKED EXAMPLE

120 mL of a 20% w/w solution, specific gravity 1.15:

weight of solution = $120 \times 1.15 = 138 \text{ g}$

drug required = $138 \times 0.20 = 27.6 \text{ g}$

23.3 Volume in volume (v/v)

For v/v the parts by volume of a liquid constituent are expressed per 100 mL of preparation.

WORKED EXAMPLE + ALCOHOL DILUTION

4 mL liquefied phenol in 250 mL lotion: $(4 \times 100)/250 = 1.6\% \text{ v/v}$.

Alcohol dilution: volume of stronger alcohol = $(\text{volume required} \times \% \text{ required}) \div \% \text{ available}$

Prepare 500 mL of 20% from 95%: $(500 \times 20)/95 = 105 \text{ mL}$ of 95% alcohol.

§24 RATIO STRENGTH, DILUTION & ALLIGATION

Very dilute preparations are often stated as a **ratio strength** — parts of solute per parts of solution, e.g. 1 in 1000. Convert to percent by dividing 100 by the second figure: 1 in 400 = $100/400 = 0.25\%$; 1 in 1000 = 0.1% ; 3 in 1000 = $300/1000 = 0.3\%$.

24.1 Solutions issued in concentrated form**FORMULA + EXAMPLE**

Strength of dilute solution = strength of concentrate $\div$ degree of dilution

1 part diluted with 7 parts water gives 1 in 8000 (= 0.0125%); dilution = $8 \times$

$\Rightarrow$ strength of concentrate = $0.0125 \times 8 = 0.1\% \text{ w/v}$

24.2 Alligation

Alligation solves problems about mixing components of different strengths. **Alligation medial** finds the weighted-average strength of a mixture of known quantities and concentrations. **Alligation alternate** finds the proportions of components of differing strengths needed to reach a desired strength, by pairing one stronger and one weaker than the target, which is placed in the centre of the working diagram.

ALLIGATION MEDIAL — zinc oxide ointment

$0.10 \times 200 \text{ g} = 20 \text{ g}$; $0.20 \times 50 \text{ g} = 10 \text{ g}$; $0.05 \times 100 \text{ g} = 5 \text{ g}$

Totals: 350 g contains 35 g $\Rightarrow 35/350 = 0.10 = 10\% \text{ w/w}$

ALLIGATION ALTERNATE — 95% and 50% alcohol to make 70%

95 part: $70 - 50 = 20$ parts of 95%

50 part: $95 - 70 = 25$ parts of 50%

$\Rightarrow$ mix in the ratio 20 : 25 (i.e. 4 : 5)

§25 WORKED PRACTICE PROBLEMS

The following representative problems, drawn from the calculation exercises, are fully worked to model the method and layout expected. Work through them with the equivalents and formulae from the preceding sections.

Units and conversions

PROBLEM 1 — capsule contents

A capsule holds 0.075 g, 20 mg, 0.0005 g, 4 mg and 500 mg. Total weight?

Convert all to mg: $75 + 20 + 0.5 + 4 + 500 = 599.5$ mg (≈ 0.5995 g).

PROBLEM 2 — dose from a concentration

Injection 50 mcg/0.5 mL. Volume for a 0.04 mg (= 40 mcg) dose?

$50 \text{ mcg} / 0.5 \text{ mL} = 40 \text{ mcg} / x \Rightarrow x = (40 \times 0.5) / 50 = 0.4 \text{ mL}$.

Specific gravity

PROBLEM 3 — weight of alcohol

Weight of 3620 mL of alcohol, specific gravity 0.82?

3620 mL water weighs 3620 g $\Rightarrow 3620 \times 0.82 = 2968.4$ g (≈ 2.97 kg).

PROBLEM 4 — specific gravity of an oil

54.96 mL of oil weighs 52.78 g; 54.96 mL water weighs 54.96 g.

specific gravity = $52.78 / 54.96 = 0.960$.

Percentage strength

PROBLEM 5 — drug content from % w/v

CLOBEX lotion 0.05% w/v in a 118 mL container. Drug content (mg)?

$0.05 \text{ g} / 100 \text{ mL} \Rightarrow (0.05 \times 118) / 100 = 0.059 \text{ g} = 59 \text{ mg}$.

PROBLEM 6 — percentage strength w/v

3 g boric acid made up to 240 mL. Strength (% w/v)?

$(3 / 240) \times 100 = 1.25\% \text{ w/v}$.

Alligation

PROBLEM 7 — mixing ointments

In what proportion mix 20% benzocaine ointment with base (0%) to get 2.5%?

20% line: $2.5 - 0 = 2.5$ parts of 20% ointment

0% line: $20 - 2.5 = 17.5$ parts of base $\Rightarrow$ ratio 2.5 : 17.5 (= 1 : 7).

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